中英对照。英文为原文，中文为译文。类名、参数名、函数名、资源路径与网址保持英文，便于在虚幻引擎细节面板中定位。插图、标题和条目顺序与原 PDF 一致。



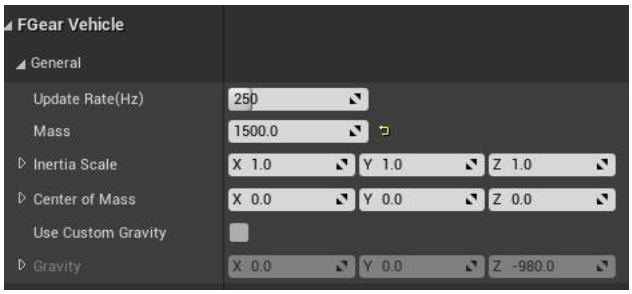
Core Modules

核心模块

If you have completed the quickstart tutorial you already have a driveable vehicle. All of the settings of that vehicle are default values so in this tutorial we will walk through all these settings and explain them briefly.
如果已经完成快速入门，你就有了一辆可驾驶的载具。该载具的全部设置都是默认值，因此本教程会逐项说明这些设置。

Vehicle Settings(General)

载具设置（General）



• Update Rate(Hz) : Vehicles require more precise calculations(*) so regular frame update rates may not be suitable for the task. This parameter sets the update frequency of the vehicles so even if the frame rate is 30 or 60hz and the update rate is 250hz then the system makes a couple of internal sub steps to make sure you will get a 250hz update rate. Keep in mind that higher rates result worse performance. Starting with v1.2 substepping is supported and it is the recommended approach. Additionally you can get a substep tick event from the skeletal mesh component to use in blueprints.

Update Rate（Hz）（更新频率）：载具需要更精确的计算（*），常规的帧更新频率可能不足以胜任。该参数设置载具的更新频率。因此即使帧率为 30 或 60 Hz，而更新频率为 250 Hz，系统也会做若干内部子步，以保证得到 250 Hz 的更新频率。请记住，频率越高性能越差。从 v1.2 起支持物理子步，并且这是推荐做法。另外可以从骨骼网格组件取得子步 Tick 事件，供蓝图使用。

• Mass : This is the mass of the vehicle body in kilograms and this value overrides the mass of the physics body. Keep it close to realistic values.

Mass（质量）：这是车身质量，单位为千克，该值会覆盖物理刚体的质量。请尽量接近真实数值。

• Inertia Scale : This value is a direct multiplier to the inertia tensor scale of the physics body. Higher values result slower accelerations and more conservation of momentum. Experiment with it as you like.

Inertia Scale（惯量缩放）：这是物理刚体惯量张量缩放的直接乘数。数值越高，加速越慢，动量保持得越多。可以按需要试验。

• Center of Mass Offset : The inertia and center of mass calculation is based on the physics asset of the vehicle. If you use the custom collision component then you can use multiple colliders with various sizes to achieve the desired effect which gives you the freedom to position center of mass as you like and usually you do not need to alter this setting. Center of mass has a dramatic effect on the handling of the vehicle so you can further tweak it with this parameter.

Center of Mass Offset（质心偏移）：惯量和质心是根据载具的物理资产计算的。如果使用自定义碰撞组件，可以用多个不同尺寸的碰撞体达到想要的效果，从而自由放置质心，通常就不必再改这项。质心对载具操控影响很大，因此还可以用该参数进一步微调。

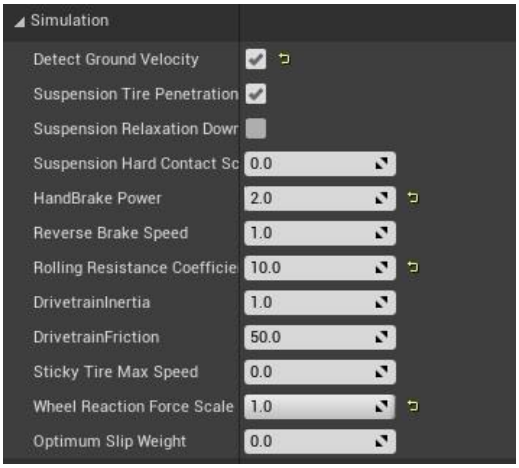
*https://www.lfs.net/forum/thread/48927-Racing-sim-physics-engine-rates
*https://www.lfs.net/forum/thread/48927-Racing-sim-physics-engine-rates

• Use Custom Gravity : If checked you can manually provide a gravity value for this vehicle.
Use Custom Gravity（使用自定义重力）：若勾选，可以手动为该载具提供重力值。

• Gravity : Gravitational acceleration vector to be used for this vehicle. You can also modify this dynamically. Check CustomGravity map for an example.
Gravity（重力）：该载具使用的重力加速度向量。也可以在运行时动态修改。示例见 CustomGravity 地图。

Vehicle Settings(Simulation)

载具设置（Simulation）



• Detect Ground Velocity : Check this if you want your vehicle to stay on moving platforms. Normally the ground velocity of the wheels are calculated just from the rigidbody but if you enable this, the velocity of the rigidbody that the wheels have contact with, will be taken into account while calculating the ground velocity. If the ground has no rigidbody then this will have no effect.
Detect Ground Velocity（检测地面速度）：如果希望载具留在移动平台上，请勾选。通常车轮的地面速度只根据刚体计算；启用后，计算地面速度时还会计入车轮所接触刚体的速度。若地面没有刚体，则没有效果。

• Suspension Tire Penetration : It is not always possible to keep the suspension compression within limits. If the suspension is compressed beyond its limits then there are two possibilities. If this is checked the tires will penetrate the ground else the tires will stay on ground but can penetrate the fenders.
Suspension Tire Penetration（悬架轮胎穿透）：并不总能把悬架压缩量限制在范围内。若悬架压缩超过极限，有两种可能。勾选后轮胎会穿透地面；不勾选则轮胎留在地面上，但可能穿透翼子板。

• Suspension Relaxation Downforce : If this is checked the relaxation damper can generate forces that pulls the vehicle towards the ground. This is mostly unnoticable at low values but if you provide a large value to relexation damper then you can observe that the vehicle will tend to stick to the ground even on bumpy surfaces. This is unrealistic but can be useful in certain cases where you need stable vehicles on rough terrain.
Suspension Relaxation Downforce（悬架回弹下压力）：若勾选，回弹阻尼可以产生把载具拉向地面的力。数值较低时大多察觉不到；若给回弹阻尼一个很大的值，可以看到载具即使在颠簸表面上也倾向贴住地面。这不真实，但在需要让载具在粗糙地形上保持稳定的某些情况下有用。

• Suspension Hard Contact Scale : If used the suspensions will generate an extra impulse after being fully compressed. This extra impulse will try to prevent the suspension from compressing more then its limit. Using a reasonable number like 0.2 is recommended for most cases, larger values can be jumpy.
Suspension Hard Contact Scale（悬架硬接触缩放）：若使用，悬架在完全压缩后会额外产生一个冲量。

该冲量试图阻止悬架压缩超过其极限。多数情况下建议使用 0.2 这类合理数值，更大的值可能造成跳动。

• Handbrake Power : Normally brake powers are set per axle but if you want the rear wheels to lock quicker with handbrake you can increase this value. This has little to no effect on actual braking power but wheels lock more easily.
Handbrake Power（手刹力度）：通常制动力按车轴设置。如果希望后轮在手刹时更快抱死，可以提高该值。它对实际制动力几乎没有影响，但车轮更容易抱死。

• Reverse Brake Speed : If the vehicles intended moving direction does not match its velocity the vehicle will automatically brake when the current speed is above this value. For ex. the vehicle is in first gear but it is moving backwards, when you press the throttle it will engage the brakes for you. The same goes for the reverse gear. This will be disabled when zero.
Reverse Brake Speed（反向制动速度）：若载具意图的行驶方向与其速度不一致，且当前速度高于该值，载具会自动制动。例如载具在一挡却正在后退，此时踩油门会替你施加制动。倒挡也是如此。设为 0 时禁用。

• Rolling Resistance Coefficient : This is the inherent friction factor of wheels. The applied friction force increases linearly as the wheels rotate faster. You may adjust it to limit the maximum speed but be aware that it also decreases the acceleration of the vehicle.
Rolling Resistance Coefficient（滚动阻力系数）：这是车轮固有的摩擦因数。车轮转得越快，施加的摩擦力线性增大。可以用来限制最高速度，但注意它也会降低载具的加速能力。

• Drivetrain Inertia : This used when calculating the acceleration/deceleration of the wheels based on applied torque. Higher inertia will result slower accelerations. This has an effect similar to wheel mass values.
Drivetrain Inertia（传动系统惯量）：根据施加的扭矩计算车轮加减速时使用该值。惯量越高，加速越慢。效果与车轮质量相近。

• Drivetrain Friction : Wheels decelerate based on effects like engine braking, rolling resistance and aerodynamic drag. This is another factor that resists the wheel rotation. Unlike rolling resistance this is a constant factor.
Drivetrain Friction（传动系统摩擦）：车轮会因发动机制动、滚动阻力和气动阻力等效果而减速。这是另一个阻碍车轮转动的因素。与滚动阻力不同，它是一个常数因子。

• Sticky Tire Max Speed : Pacejka tire model does not work well at low speeds. When the velocity gets too small the calculations get problematic so tires can slide slowly on a slope even if the vehicle is fully stopped. This is an experimental hacky solution to solve this issue. When the vehicles velocity is below this speed(kmh) the tires will switch to sticky state. If you keep this value at 0 then it is disabled. The effect of this parameter depends on substepping and you may need to use a larger value if substepping is off.
Sticky Tire Max Speed（低速黏附最高速度）：Pacejka 轮胎模型在低速时表现不好。速度过小时计算会出问题，因此即使载具已经完全停住，轮胎仍可能在斜坡上缓慢滑动。这是一个试验性的权宜办法，用来解决该问题。载具速度低于该值（km/h）时，轮胎切换到黏附状态。保持为 0 则禁用。该参数的效果依赖物理子步；若关闭子步，可能需要使用更大的值。

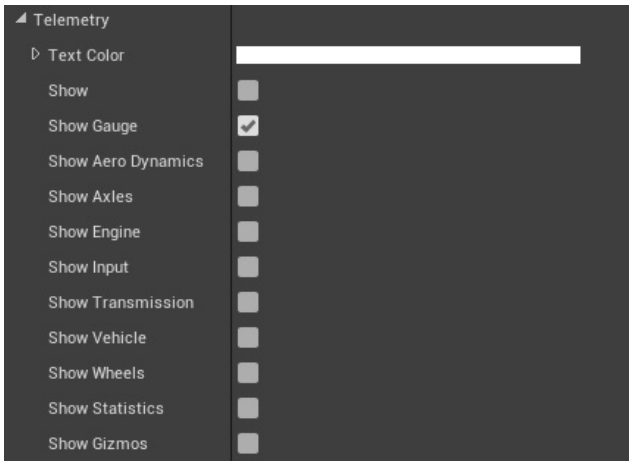
• Wheel Reaction Force Scale : Tires generate friction forces to move the vehicle and suspensions generate forces to keep the vehicle away from ground. If you want the generated forces applied back to dynamic objects in your scene set this variable to 1. Sometimes generated forces are too large for objects with low mass so you can use a smaller value instead.
Wheel Reaction Force Scale（车轮反作用力缩放）：轮胎产生摩擦力来推动载具，悬架产生力使载具离开地面。如果希望这些力反向作用到场景中的动态物体上，把该变量设为 1。有时产生的力对低质量物体过大，这时可以使用更小的值。

• Optimum Slip Weight : This is an experimental feature, use with caution. Optimum slip is the value at which the tire generates the greatest friction force. This basically adjusts the slip ratio to prevent it from going below the optimum value. The result is that the engine rpm and the wheel speeds will be at perfect match but having a constant slip value can effect the overall lateral traction too.

Optimum Slip Weight（最佳滑移权重）：这是试验功能，请谨慎使用。最佳滑移是轮胎产生最大摩擦力时的滑移值。它基本上会调整滑移率，防止其降到最佳值以下。结果是发动机转速与车轮转速会完全匹配，但恒定的滑移值也会影响整体侧向抓地。

Vehicle Settings(Telemetry)

载具设置（Telemetry）

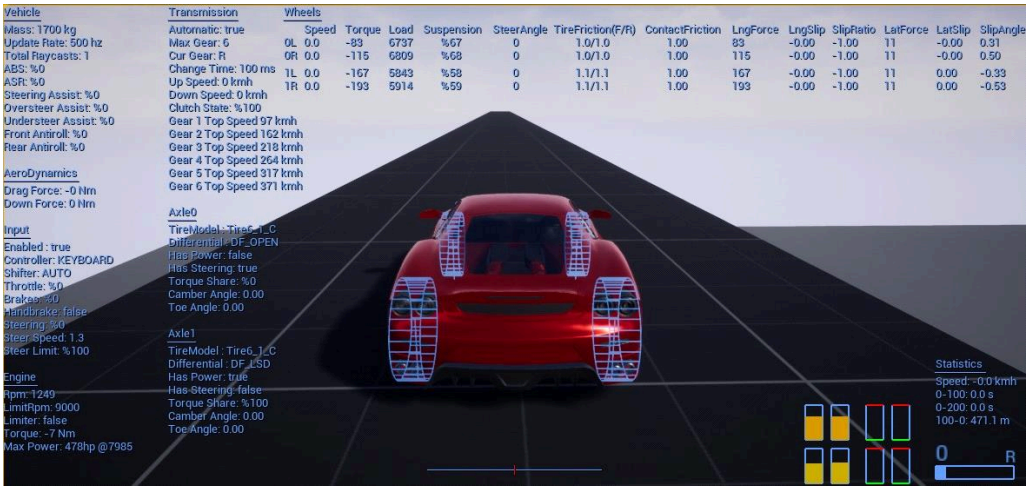


• Text Color : Text color of telemetry ui.

Text Color（文字颜色）：遥测界面的文字颜色。

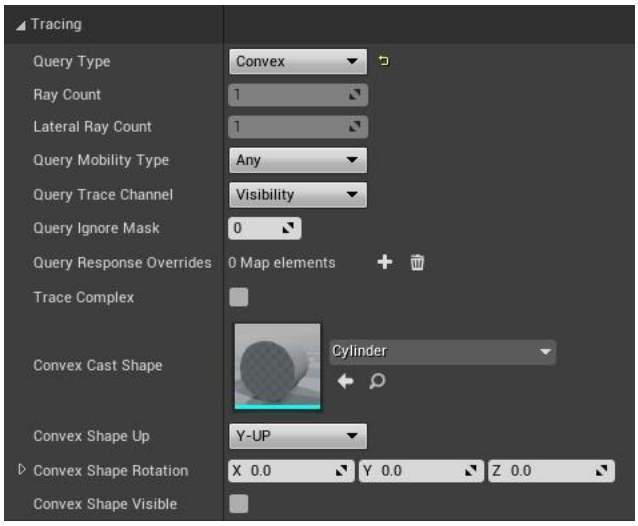
• Show : Shows/Hides all telemetry windows available. Use other checkboxes for each individual window to show or hide. A sample of all telemetry shown below.

Show（显示）：显示或隐藏全部可用的遥测窗口。用其他复选框分别显示或隐藏每个窗口。下方是全部遥测同时显示的示例。



Vehicle Settings(Tracing)

载具设置（Tracing）

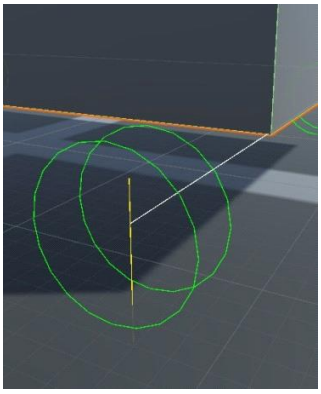


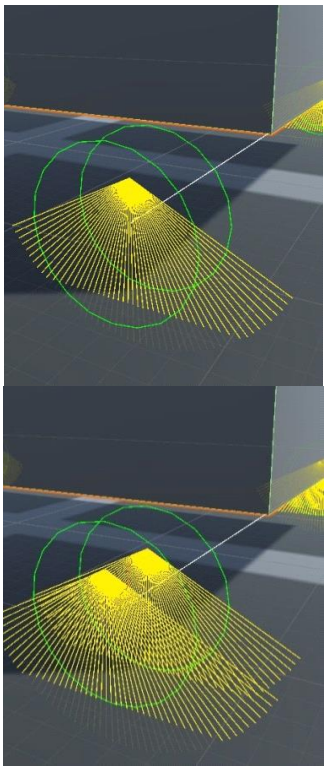
• Query Type : This is the type of raycasting that is used to find the tire contact position. By default raycasting is used. Sphere casting gives a nice 3d wheel effect but the downside is that your wheels will behave like a sphere instead of a cylinder. Convex casting lets you choose a convex collider to be casted. You can specify a custom mesh for each wheel in wheel settings. If you do not specify a mesh collider the default cylinder model in "FGearPlugin/Content/Mesh/Cylinder" is used. Cylinder casting is smooth like sphere casting but not as precise since convex colliders have limited triangle counts. If you want to provide a custom mesh for casting, make sure to set its transform params properly. Warning : Convex casting is more sensitive to floating point precision issues so for ex. if you put a very large plane below the vehicle, casts can return unexpected results and vehicles can shake wildly for no reason.

Query Type（查询类型）：用于寻找轮胎接触位置的射线检测类型。默认使用射线检测。球体投射能给出不错的三维车轮效果，但缺点是车轮会表现得像球体而不是圆柱。凸体投射允许选择一个要投射的凸碰撞体。可以在车轮设置中为每个车轮指定自定义网格。若不指定网格碰撞体，则使用 "FGearPlugin/Content/Mesh/Cylinder" 中的默认圆柱模型。圆柱投射像球体投射一样平滑，但因为凸碰撞体的三角形数量有限，精度不如射线。若要提供自定义投射网格，请正确设置其变换参数。警告：凸体投射对浮点精度问题更敏感。例如在载具下方放一个非常大的平面时，投射可能返回意外结果，载具会无故剧烈抖动。

• Ray Count : This represents the number of rays that are casted for each wheel. If you need a 3d wheel effect you can increase this value. Keep in mind that the higher number of raycasts, the worse performance you get. For a racing game its better to keep its value at 1.

Ray Count（射线数量）：每个车轮投射的射线数量。如果需要三维车轮效果可以提高该值。请记住，射线越多性能越差。竞速游戏最好保持为 1。





• Lateral Ray Count : If you need a wheel with real width than setting this value to 2 or 3 will multiply the Ray Count value and the origins of raycasts are offset to left and right according to the wheel width. For a racing game its better to keep its value at 1.
Lateral Ray Count（横向射线数量）：如果需要具有真实宽度的车轮，把该值设为 2 或 3 会乘上 Ray Count，并且射线起点会按轮宽向左右偏移。竞速游戏最好保持为 1。

• Query Mobility Type : Select to query against static or dynamic or both.
Query Mobility Type（查询可动性类型）：选择对静态、动态或两者进行查询。

• Query Trace Channel : This is the collision channel parameter of LineTraceSingleByChannel function*.
Query Trace Channel（查询追踪通道）：LineTraceSingleByChannel 函数的碰撞通道参数*。

• Query Ignore Mask : Extra filtering parameter of type FMaskFilter. Quoting unreal engine source comment : "This filter allows us to refine queries (channel, object) with an additional level of ignore by tagging entire classes of objects..."
Query Ignore Mask（查询忽略遮罩）：类型为 FMaskFilter 的额外过滤参数。引用虚幻引擎源码注释："该过滤器允许我们通过给整类对象打标签，为查询（通道、对象）增加一级忽略……"

• Query Response Overrides : This only works with line & sphere tracing. By default all channel responses are set to blocked, if you want to modify a channel response you can add an override to this list.
Query Response Overrides（查询响应覆盖）：仅对线段和球体追踪有效。默认所有通道响应都设为阻挡。如果要修改某个通道的响应，可以向该列表添加一条覆盖。

• Trace Complex : Whether we should trace against complex(mesh) collision.
Trace Complex（追踪复杂碰撞）：是否对复杂（网格）碰撞进行追踪。

• Convex Cast Shape : This is only enabled if you choose convex casting. If you do not specify a mesh collider the default cylinder model in "FGearPlugin/Content/Mesh/Cylinder" is used. If you want to provide a custom mesh for casting, make sure it's size and orientation matches the sample asset or you need to play with the transform parameters below.
Convex Cast Shape（凸体投射形状）：仅在选择凸体投射时启用。若不指定网格碰撞体，则使用 "FGearPlugin/Content/Mesh/Cylinder" 中的默认圆柱模型。若要提供自定义投射网格，请确保其尺寸和朝向与示例资产一致，否则需要调整下面的变换参数。

• Convex Shape Up : By default(Y-Up) the convex casting mesh is scaled with the radius on X and Z axis and with the width on Y axis. If you provide a custom convex mesh, this parameter lets you to change how the

mesh is scaled.
Convex Shape Up（凸体形状向上轴）：默认（Y-Up）时，凸体投射网格在 X 轴和 Z 轴上按半径缩放，在 Y 轴上按宽度缩放。若提供自定义凸网格，该参数用来改变网格的缩放方式。

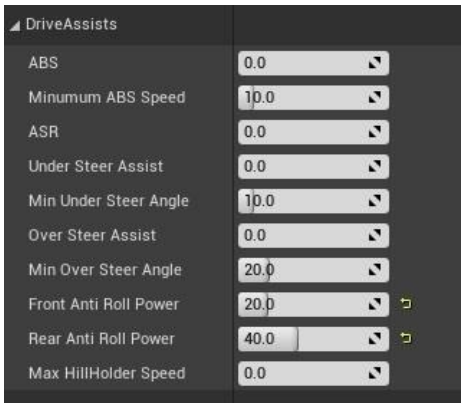
• **Convex Shape Rotation : Sets the local rotation of the convex casting mesh.**
Convex Shape Rotation（凸体形状旋转）：设置凸体投射网格的局部旋转。

• **Convex Shape Visible : If set, the convex casting mesh is displayed on its default position, this is useful for debugging while tuning the casting mesh.**
Convex Shape Visible（显示凸体形状）：若设置，凸体投射网格会显示在其默认位置，便于在调整投射网格时调试。

***https://docs.unrealengine.com/en-US/Engine/Physics/Tracing/HowTo/SingleLineTraceByChannel/index.html**
*https://docs.unrealengine.com/en-US/Engine/Physics/Tracing/HowTo/SingleLineTraceByChannel/index.html

Vehicle Settings(DriveAssists)

载具设置（DriveAssists）



• **ABS : The antilock braking system helps to prevent wheels from locking up when braking.**
ABS：防抱死制动系统，有助于防止制动时车轮抱死。

• **Minumum ABS Speed : The antilock braking system will not be active below this speed.**
Minumum ABS Speed（ABS 最低速度）：低于该速度时，防抱死制动系统不会工作。

• **ASR : Traction control assist helps to prevent loss of traction of driven wheels when accelerating.**
ASR：驱动防滑辅助，有助于防止驱动轮在加速时失去抓地。

• **Under Steer Assist : Helps to prevent understeering by using brakes like an esp system. This only works for 4 wheel vehicles.**
Under Steer Assist（转向不足辅助）：像 ESP 系统那样使用制动来防止转向不足。仅对四轮载具有效。

• **Min UnderSteer Angle : If the angle between the steering direction and the vehicles velocity direction is above this value, under steer assist is activated.**
Min UnderSteer Angle（最小转向不足角）：若转向方向与载具速度方向之间的夹角高于该值，则启用转向不足辅助。

• **Over Steer Assist : Helps to prevent oversteering by using brakes like an esp system. This only works for 4 wheel vehicles.**
Over Steer Assist（转向过度辅助）：像 ESP 系统那样使用制动来防止转向过度。仅对四轮载具有效。

• **Min OverSteer Angle : If the angle between the steering direction and the vehicles velocity direction is above this value, over steer assist is activated.**
Min OverSteer Angle（最小转向过度角）：若转向方向与载具速度方向之间的夹角高于该值，则启用转向过度辅助。

• **Front AntiRoll : Anti roll bars help to reduce body roll during fast cornering. Front antirollbars help to reduce oversteering. This only works for 4 wheel vehicles. Current implementation of antiroll bars transfer**

spring force between wheels which is kinda realistic but as the values get close to %100 instabilities may occur so try to keep it around %50. There is also an alternative(unrealistic but effective) implementation in ArcadeAssists.cs script.
Front AntiRoll（前防倾）：防倾杆有助于减小快速过弯时的车身侧倾。前防倾杆有助于减小转向过度。仅对四轮载具有效。当前实现是在车轮之间传递弹簧力，比较接近真实情况，但数值接近 100% 时可能不稳定，因此尽量保持在 50% 左右。ArcadeAssists.cs 脚本中还有另一种（不真实但有效的）实现。

• **Rear AntiRoll : Rear anti rollbars help to reduce understeering. This only works for 4 wheel vehicles.**
Rear AntiRoll（后防倾）：后防倾杆有助于减小转向不足。仅对四轮载具有效。

• **Max Hill Holder Speed : Hill-holder holds the brake when the vehicle travel direction is against the current gear. It's only activated if the speed of the vehicle is below this value. Zero means disabled.**
Max Hill Holder Speed（坡道保持最高速度）：当载具行驶方向与当前挡位相反时，坡道保持会握住制动。仅当载具速度低于该值时启用。零表示禁用。

Vehicle Settings(Optimization)

载具设置（Optimization）



• **Sleep Waiting Time : Automatic sleeping time. When this is positive the system checks the vehicles linear and angular velocities and if they’re below defined thresholds then the vehicle is put to sleep.**
Sleep Waiting Time（休眠等待时间）：自动休眠时间。该值为正时，系统检查载具的线速度和角速度，若都低于设定阈值，则让载具进入休眠。

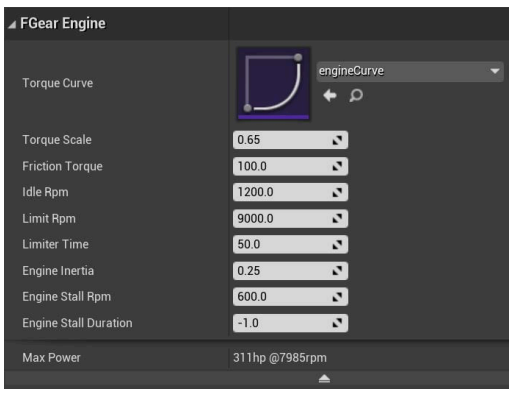
• Sleep Linear Threshold : In order to put the vehicle to sleep its linear speed should be lower then this value. The unit is kilometers per hour.
Sleep Linear Threshold（休眠线速度阈值）：要让载具休眠，其线速度应低于该值。单位是千米每小时。

• Sleep Angular Threshold : In order to put the vehicle to sleep its angular speed should be lower then this value. The unit is degrees per second.
Sleep Angular Threshold（休眠角速度阈值）：要让载具休眠，其角速度应低于该值。单位是度每秒。

• Wake Up Sources : Defines which inputs can wake the vehicle up.
Wake Up Sources（唤醒来源）：定义哪些输入可以把载具唤醒。

Engine Settings

发动机设置



• **Torque Curve : The engine torque is the result of this curve scaled with the scale parameter. The X axis of the curve represents engine rpm and Y axis is the torque value in Nm. There are real world vehicles engine curves available on the internet, copying those curves is a good way to start.**
Torque Curve（扭矩曲线）：发动机扭矩是该曲线的结果再乘以缩放参数。曲线 X 轴表示发动机转速，

Y 轴是扭矩，单位为牛·米。网上可以找到真实车辆的发动机曲线，从抄录这些曲线开始是个好办法。

• Torque Scale : This is the direct multiplier of engine torque curve. Final engine torque is : Curve Result x Scale. Normally you would keep this as is but you can alter it for temporary engine boosts.
Torque Scale（扭矩缩放）：发动机扭矩曲线的直接乘数。最终发动机扭矩为：曲线结果 × 缩放。通常保持不变，但可以临时提高以获得发动机增益。

• Friction Torque : This is a constant friction factor(multiplied with rpm) that slows down the engine. Higher values result more engine brake and faster rpm rev downs. Lower values has opposite effect so to determine a good value you can rev up in neutral gear and see how it revs down.
Friction Torque（摩擦扭矩）：一个恒定的摩擦因数（会乘以转速），用来使发动机减速。数值越高，发动机制动越强，转速下降越快。数值越低则相反。要确定一个合适的值，可以在空挡把转速拉高，再看它如何回落。

• Idle Rpm : Minumum rpm value of the engine, currently there is no stalling so the engine always keeps revving.
Idle Rpm（怠速）：发动机的最低转速。当前没有熄火，因此发动机会一直保持运转。

• Limit Rpm : This is the maximum rpm that the engine can rev up to, after this point engine input is blocked for a little time, see limiter time.
Limit Rpm（限制转速）：发动机能够拉到的最高转速。超过该点后，发动机输入会短暂被阻断，见限制器时间。

• Limiter Time : When the rpm value reaches the limit value the throttle input is blocked for this long. It is measured in miliseconds. Experiment with it but keep in mind that if the engine inertia is too low, engine rpm will fluctuate a lot so you better keep it as low as possible.
Limiter Time（限制器时间）：转速达到限制值时，油门输入会被阻断这么长时间。单位为毫秒。可以试验，但请记住，若发动机惯量过低，转速会大幅波动，因此最好尽量保持较小。

• Engine Inertia : This effects the acceleration/deceleration of the engine. Inertia has direct effect on the acceleration of the vehicle so avoid too low or high values. To determine a good value you can rev up/down in neutral gear and see how it behaves.
Engine Inertia（发动机惯量）：影响发动机的加减速。惯量直接影响载具的加速，因此避免过低或过高。要确定合适的值，可以在空挡升降转速并观察表现。

• Engine Stall Rpm : Engine is considered to be stalling when the current rpm falls below this value. You could use around 50-90 percent of the idle rpm value.
Engine Stall Rpm（发动机熄火转速）：当前转速降到该值以下时，认为发动机正在熄火。可以使用怠速的大约 50% 到 90%。

• Engine Stall Duration : If the engine keeps stalling this amount of time in seconds then the engine is forced to stop. The engine will not stall if this value is negative.
Engine Stall Duration（发动机熄火持续时间）：若发动机持续熄火达到这么多秒，就会被强制停止。该值为负时发动机不会熄火。

Transmission Settings

变速箱设置



• Auto Change : When set to true gears are changed automatically according to the gear up/down values.
Auto Change（自动换挡）：设为 true 时，按升挡/降挡值自动换挡。

• Auto Reverse : When set to true the transmission automatically goes to rear gear when you brake and stop. Also the same thing happens when you are going backwards and push the throttle, this time the transmission automatically goes to first gear. This only happens when Auto Change is set.
Auto Reverse（自动倒挡）：设为 true 时，制动并停住后变速箱会自动进入倒挡。倒车时再踩油门也会自动回到一挡。仅在启用 Auto Change 时发生。

• Auto Clutch : When set to true the transmission automatically manages the clutch input for you. Set this to false if you want to manually operate the clutch.
Auto Clutch（自动离合器）：设为 true 时，变速箱会自动管理离合器输入。如果要手动操作离合器，设为 false。

• Rev Matching : When set to true the rpm of the engine is interpolated towards the perfect match of your next gear ratio while changing gears. The time to interpolate is based on change time.
Rev Matching（转速匹配）：设为 true 时，换挡过程中发动机会向下一挡传动比的理想匹配转速插值。插值时间基于换挡时间。

• Change Time : When a gear change happens first the transmission goes to neutral and then switches to the next gear, this value determines how long this process takes in miliseconds.
Change Time（换挡时间）：换挡时变速箱先进入空挡，再切到下一挡。该值决定这个过程持续多少毫秒。

• Gear Up Ratio : Suppose you drive in 1st gear, the maximum speed that you can achieve with this gear is calculated(mxs) and if you go over (gearup ratio * mxs) then the gear is changed.
Gear Up Ratio（升挡比）：假设在一挡行驶，先算出该挡能达到的最高速度（mxs），若超过（升挡比 × mxs）就换挡。

• Gear Down Ratio : Suppose you drive in 2nd gear, the maximum speed that you can achieve with the previous gear(1st) is calculated(mxs) and if you go below (geardown ratio * mxs) then the gear is changed.
Gear Down Ratio（降挡比）：假设在二挡行驶，先算出前一挡（一挡）能达到的最高速度（mxs），若低于（降挡比 × mxs）就换挡。

• Gear Up Velocity Bias : Gear change can be based on vehicles velocity or the wheel speed. If this is zero it will be %100 based on wheel speed and if this is 1 then it will be %100 based on velocity.
Gear Up Velocity Bias（升挡速度偏向）：换挡可以基于载具速度或车轮转速。为 0 时 100% 基于车轮转

速，为 1 时 100% 基于载具速度。

• Gear Down Velocity Bias : Same as above but this time it is for shifting down.
Gear Down Velocity Bias（降挡速度偏向）：与上一项相同，但用于降挡。

• Clutch Engage Time : Before changing gears, the clutch is fully disengaged. After gear change is completed, it takes this amount of time to reengage the clutch. Usually this value should be equal or lower than the change time.
Clutch Engage Time（离合器接合时间）：换挡前离合器完全分离。换挡完成后，需要这么长时间重新接合离合器。通常该值应等于或小于换挡时间。

• Clutch Power Scale (%) : This is a coefficient to the power transfer of the clutch(between the engine and the wheels). Do not change it if you really need to. Lower values will result high differences between engine and wheel rpms. Higher values will force both rpm values to be in sync but the engine rpm can behave erratic.
Clutch Power Scale（%）（离合器动力缩放）：离合器（发动机与车轮之间）动力传递的系数。除非确实需要，否则不要修改。较低的值会使发动机转速与车轮转速差异很大。较高的值会迫使两者同步，但发动机转速可能变得不稳定。

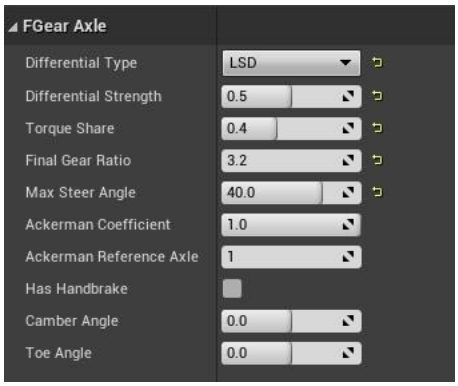
• Gear Ratios : This is an array of values which are used for calculating the final transmission ratio along with final drive. If you need a 5 speed car then the size should be 6 with the addition of rear gear. So the first 5 values are forward ratios and the last one is the rear gear ratio. All values should be positive and you can find real cars values on the internet. The default vehicle has only 1 forward and 1 backward gear so you can try the values in the image above.
Gear Ratios（齿比）：一组数值，与主减速比一起用来计算最终传动比。若需要 5 前进挡，则数组大小应为 6，再加上倒挡。因此前 5 个是前进齿比，最后一个是倒挡齿比。所有值都应为正，网上可以找到真实车辆的数值。默认载具只有 1 个前进挡和 1 个倒挡，可以试上图中的数值。

• Stall Assist : This is an experimental feature. When you first launch the vehicle or when you try to climb a steep slope, the rpm can stay at a low level until you reach a certain speed since there is not enough torque and there is no engine stalling. This is quite observable when you have a long first fear ratio. This assist will scale the power of the clutch in order to fight with this behavior. With lower clutch power the rpm can rise up and you can get receive more torque from the engine resulting faster accelerations. This is only activated with the first(1) and the rear(R) gears.
Stall Assist（熄火辅助）：这是试验功能。起步或爬陡坡时，由于扭矩不足且没有发动机熄火，转速可能一直偏低，直到达到某个速度。一挡传动比较长时这一点很明显（原文写作 first fear ratio）。该辅助会缩放离合器的动力传递来对抗这种表现。离合器动力较低时转速可以拉高，从而从发动机得到更多扭矩，加速更快。仅在一挡（1）和倒挡（R）时启用。

Clutch Scale Min(%) : The minumum clutch power we can fall down to. Clutch Scale Max(%) : The maximum clutch power we can rise up to.
Clutch Scale Min（%）（离合器缩放下限）：离合器动力能够降到的最小值。Clutch Scale Max（%）（离合器缩放上限）：离合器动力能够升到的最大值。

Axle Settings
车轴设置

This project is currently focused on 2 axled cars but you may add additional axles if you want*. Axles are indexed/named with numbers starting with 0.
本工程目前针对两轴汽车，但也可以按需要添加更多车轴*。车轴从 0 开始用数字索引和命名。



• Differential Type : If you do not know what a differential is check some videos. Currently open, locked and lsd differentials are supported.**
Differential Type（差速器类型）：如果不知道差速器是什么，可以看一些视频**。当前支持开放式、锁止式和限滑差速器。

• Differential Strength : This parameter is only available if lsd type is selected and determines the magnitude of the extra force that is applied to the gripping wheel.
Differential Strength（差速器强度）：仅在选择限滑类型时可用，决定施加到有抓地力车轮上的额外力的大小。

• Torque Share : The ratio of the torque received from engine. In reality the sum of the shares should be 1 but it is not constraint.
Torque Share（扭矩份额）：从发动机得到的扭矩比例。现实中各份额之和应为 1，但这里并不强制。

• Final Gear Ratio : The transmission value is calculated from the current gear ratio and this final gear ratio(also called final drive). In real cars this is the gear ratio between the driveshaft and the axle which can determine the acceleration and the top speed of the vehicle. Higher values result faster accelerations and lower top speeds. You only need to set this for powered axles. The gear ratios should match the wheel radius ratios so if the front wheels are smaller than the final gear should be larger etc.
Final Gear Ratio（最终传动比）：传动值由当前挡位齿比和该最终传动比（也叫主减速比）计算。在真实汽车中，这是传动轴与车轴之间的齿比，可以决定加速和最高速度。数值越高加速越快、最高速度越低。只需为有动力的车轴设置。齿比应与车轮半径比相匹配，因此若前轮更小，最终传动比就应更大，依此类推。

• Max Steer Angle : Maximum angle(degree) of steering for this axles wheels. For non steering wheels just leave it at 0.
Max Steer Angle（最大转向角）：该车轴车轮的最大转向角（度）。非转向轮保持为 0。

• Ackerman Coefficient : Ackerman steering values are automatically calculated from this axle and the reference axle. This value determines the rate this calculation is applied. Providing negative values for anti ackerman is also possible.
Ackerman Coefficient（阿克曼系数）：阿克曼转向值由该车轴和参考车轴自动计算。该值决定这一计算被应用的比例。也可以提供负值来做反阿克曼。

• Ackerman Reference Axle : Ackerman calculation is based on the current axle and the axle that is referenced by this index value. Normally this value is the index of the rear axle.
Ackerman Reference Axle（阿克曼参考车轴）：阿克曼计算基于当前车轴和该索引所引用的车轴。通常该值是后轴的索引。

• Has Handbrake : If set this axles wheels are effected by handbrake.
Has Handbrake（有手刹）：若设置，该车轴的车轮受手刹影响。

• Camber Angle : Camber angle value in degrees. This value visually changes the wheels roll angle and is fed to the tire model. To get a better idea of camber and toe check some videos*.**
Camber Angle（外倾角）：外倾角，单位为度。该值会在视觉上改变车轮的侧倾角，并送入轮胎模型。要更好理解外倾和前束，可以看一些视频***。

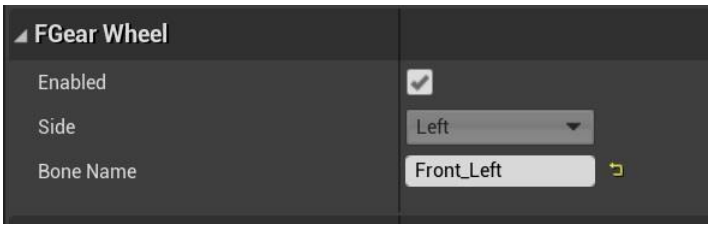
• **Toe Angle : Toe angle value in degrees. This value visually change the wheels yaw angle and modifies the direction of the force output of the wheel.**
Toe Angle（前束角）：前束角，单位为度。该值会在视觉上改变车轮的偏航角，并修改车轮力输出的方向。

* Add new FGearAxle component then add the following three child components : FGearWheelOptions and 2 x FGearWheel(left/right).
* 添加新的 FGearAxle 组件，然后添加下面三个子组件：FGearWheelOptions，以及 2 个 FGearWheel（左/右）。

**https://www.youtube.com/watch?v=yYAw79386WI
**https://www.youtube.com/watch?v=yYAw79386WI

***https://www.youtube.com/watch?v=_D_vg8gnMms
***https://www.youtube.com/watch?v=_D_vg8gnMms

Wheel Settings
车轮设置



• **Enabled : You can check/uncheck this to enable/disable a wheel. The effect is only physical so if you want to hide the disabled wheel you need to do it yourself.**
Enabled（启用）：可以勾选或取消以启用或禁用一个车轮。效果只作用于物理，因此若要隐藏被禁用的车轮，需要自己处理。

• **Side : Each axle must have two wheels and this parameter tells if this is the left or the right wheel. You should only set this parameter when you first create the wheels.**
Side（侧）：每个车轴必须有两个车轮，该参数说明这是左轮还是右轮。只应在首次创建车轮时设置。

• **Bone Name : Name of the bone that correspons to this wheel. This is used to find the initial position of the wheel. It will be positioned and rotated according to the simulation.**
Bone Name（骨骼名）：对应该车轮的骨骼名称。用于找到车轮的初始位置。之后会按模拟结果定位和旋转。

Wheel Options Settings(Tire)
车轮选项设置（Tire）



• **Tire Model : This is a reference to the TireModel class that is used for calculating wheel forces. This list will only include the tiremodel classes of default tires and additional tire model classes that you derive from. If you want to create a new tire model, create a new blueprint class and derive from a tire model class (ex.**

FGearTire96) and then you can alter its variables.
Tire Model（轮胎模型）：指向用于计算车轮力的 TireModel 类。该列表只包含默认轮胎的轮胎模型类，以及你所派生的额外轮胎模型类。若要新建轮胎模型，新建一个蓝图类并从某个轮胎模型类派生（例如 FGearTire96），然后就可以修改其变量。

• Lateral Friction : The output of the tire models lateral force is multiplied with this value. Even if you use large values the final tire force will be limited. If you want to directly adjust resulting forces then you should use the scaler parameters above.
Lateral Friction（横向摩擦）：轮胎模型横向力的输出会乘以该值。即使使用很大的值，最终轮胎力仍会受到限制。若要直接调整结果力，应使用上面的缩放参数。

• Longitudinal Friction : The output of the tire models longitudinal force is multiplied with this value. Even if you use large values the final tire force will be limited. If you want to directly adjust resulting forces then you should use the scaler parameters above.
Longitudinal Friction（纵向摩擦）：轮胎模型纵向力的输出会乘以该值。即使使用很大的值，最终轮胎力仍会受到限制。若要直接调整结果力，应使用上面的缩放参数。

• Tire Force/Moment Scalers : These scale parameters are direct multipliers to the resulting forces so using values other than "1" is out of realism. All tire models generate Fx, Fy and My. Mx and Mz are only generated by MF6.1 tire model.
Tire Force/Moment Scalers（轮胎力/力矩缩放）：这些缩放参数是结果力的直接乘数，因此使用 "1" 以外的值就超出了真实范围。所有轮胎模型都会生成 Fx、Fy 和 My。Mx 和 Mz 只由 MF6.1 轮胎模型生成。

• (Fx)Longitudinal Force Scale: The final longitudinal tire force is multiplied with this.
(Fx) Longitudinal Force Scale（纵向力缩放）：最终纵向轮胎力乘以该值。

• (Fy)Lateral Force Scale: The final lateral tire force is multiplied with this.
(Fy) Lateral Force Scale（横向力缩放）：最终横向轮胎力乘以该值。

• (Mx)Over-Turn Moment Scale: The final X-Axis moment is multiplied with this.
(Mx) Over-Turn Moment Scale（侧翻力矩缩放）：最终 X 轴力矩乘以该值。

• (My)Wheel-Roll Moment Scale: The final Y-Axis moment is multiplied with this.
(My) Wheel-Roll Moment Scale（车轮滚转力矩缩放）：最终 Y 轴力矩乘以该值。

• (Mz)Self-Align Moment Scale: The final Z-Axis moment is multiplied with this.
(Mz) Self-Align Moment Scale（回正力矩缩放）：最终 Z 轴力矩乘以该值。

• Tire Pressure : Inflation pressure of tire in bars. The default value 2 bars is about 29 psi. This is only used by MF6.1 tire model.
Tire Pressure（胎压）：轮胎充气压力，单位为巴。默认值 2 巴大约是 29 psi。仅 MF6.1 轮胎模型使用。

Wheel Options Settings(Wheel)
车轮选项设置（Wheel）



• Radius : Radius of the wheel in meters. You can see the resulting circle in play mode if you enable telemetry gizmos.
Radius（半径）：车轮半径，单位为米。若启用遥测 Gizmo，可以在运行时看到对应的圆。

• **Width : Wheels do not have real width in this system but the width(in meters) parameter is integrated to effect one thing. If you set lateral ray count to 2 or 3 then the distance between the rays are calculated using the width value. The best practice is to use realistic values and you can preview the tire width with the gizmos drawn.**
Width（宽度）：本系统中车轮并没有真实宽度，但宽度（米）参数只影响一件事。若把横向射线数量设为 2 或 3，射线间距会用宽度值计算。最好使用真实数值，并且可以用绘制的 Gizmo 预览胎宽。

• **Forward Offset : A positive offset will move the wheels towards the pointing direction of the vehicle.**
Forward Offset（前向偏移）：正偏移会把车轮沿载具指向方向移动。

• **Spacer Offset : A positive spacer offset will move the wheels away from each other resulting a wider track.**
Spacer Offset（轮距垫片偏移）：正的垫片偏移会把左右车轮彼此推开，从而加宽轮距。

• **RimOffset : This is a lateral offset to the wheel rim. If your wheel position does not match with the physical wheel position you can adjust by using this value. You can activate gizmos from telemetry settings to see if your physical wheels match your wheel models.**
RimOffset（轮辋偏移）：车轮轮辋的横向偏移。若车轮视觉位置与物理车轮位置不符，可以用该值调整。可以从遥测设置打开 Gizmo，查看物理车轮是否与车轮模型对齐。

• **CastOffset : This is an offset along the suspension normal. Use a positive value to offset the origin of ray/shape casts. If your wheels get stuck under the ground when it falls from height then you can try using this. Do not use a large value, try to keep the cast origins inside the vehicle hull. You can view the ray origins with the telemetry gizmo.**
CastOffset（投射偏移）：沿悬架法线的偏移。用正值偏移射线或形状投射的起点。若载具从高处落下后车轮卡在地面以下，可以尝试使用它。不要用很大的值，尽量让投射起点留在车身外壳内部。可以用遥测 Gizmo 查看射线起点。

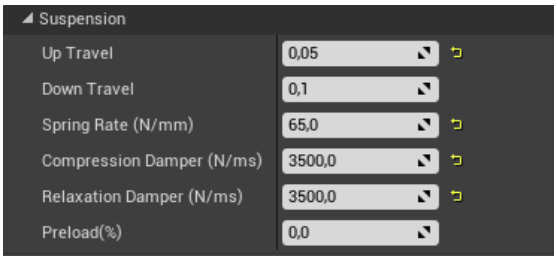
• **Mass : Represents the mass of the wheel and used for calculating the rotational acceleration of the wheels. Lower mass wheels tend to spin more and higher mass wheels spin less. Too small values usually result jittery behaviour and too high values limit accelerations so try to experiment and find logical values.**
Mass（质量）：表示车轮质量，用于计算车轮的转动加速度。质量较低的车轮更容易打滑空转，质量较高的车轮更不容易空转。过小通常会导致抖动，过大则会限制加速，因此应试验并找到合理数值。

• **Brake Torque : Brake torque applied when the braking input is %100. If you need more torque for handbrakes, there is a parameter for that in General Settings.**
Brake Torque（制动扭矩）：制动输入为 100% 时施加的制动扭矩。如果手刹需要更大扭矩，General 设置中有对应参数。

Warning : If you modify a variable(from code or blueprint) that is found in the FGearWheelOptions class, the change will not be in effect until you call FGearAxle::applyWheelOptions method. Check the Modification maps blueprint for an example.
警告：如果修改了 FGearWheelOptions 类中的变量（通过代码或蓝图），在调用 FGearAxle::applyWheelOptions 之前，改动不会生效。示例见 Modification 地图的蓝图。

Wheel Options Settings(Suspension)

车轮选项设置（Suspension）



• **Up Travel : Total suspension spring length is divided into up&down travel values. This is the maximum distance in meters that your suspension will compress after the default pose. This will not effect the visible ride height.**

Up Travel（上行程）：悬架弹簧总长度分为上行程和下行程。这是相对默认姿态，悬架能够压缩的最大距离，单位为米。它不会影响可见的行驶高度。

• Down Travel : This is the maximum distance in meters that your suspension will extend after the default pose. This value is the one that will effect the visible ride height.
Down Travel（下行程）：相对默认姿态，悬架能够伸长的最大距离，单位为米。影响可见行驶高度的是该值。

• Spring Rate : Suspension spring force per mm, total force is SpringRate x TotalLength. If your spring rate is not sufficient then you can not get the desired ride height so you are recommended to open telemetry gui and tweak accordingly.
Spring Rate（弹簧刚度）：每毫米悬架弹簧力，总力为 Spring Rate × 总长度。若弹簧刚度不足，就得不到想要的行驶高度，因此建议打开遥测界面再调整。

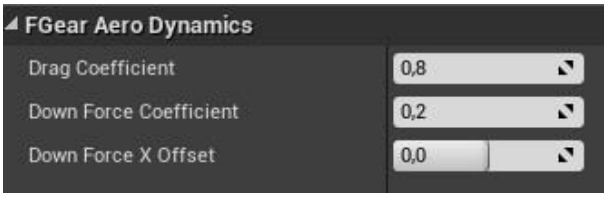
• Compression Damper: Suspension compression damper force per 1 ms velocity. This value is used when suspension is compressing and it absorbs some of the spring force and reduces suspension oscillations.
Compression Damper（压缩阻尼）：悬架压缩阻尼力，对应每 1 毫秒的速度。悬架压缩时使用该值，它会吸收一部分弹簧力并减小悬架振荡。

• Relaxation Damper : Suspension relaxation damper force per 1 ms velocity. This value is used when suspension is compressing and it absorbs some of the spring force and reduces suspension oscillations. Also see Suspension Relaxation Downforce.
Relaxation Damper（回弹阻尼）：悬架回弹阻尼力，对应每 1 毫秒的速度。悬架压缩时使用该值，它会吸收一部分弹簧力并减小悬架振荡。另见 Suspension Relaxation Downforce。

• Preload : This is the percentage of suspension length that is already compressed while it is fully extended.
Preload（预载）：悬架完全伸长时，已经被压缩的悬架长度百分比。

Suspensions have a great effect of your vehicles driving behaviour. To get a better idea of suspensions check some videos : https://www.youtube.com/watch?v=e_EAWKGvSp0
悬架对载具驾驶表现影响很大。要更好理解悬架，可以看一些视频：https://www.youtube.com/watch?v=e_EAWKGvSp0

AeroDynamics Settings
空气动力学设置



• Drag Coefficient : Aero drag is calculated using a unit area factor, vehicle speed and air density constant. This value is multiplied with the result of this calculation. The direction of the force is always the opposite of vehicles velocity.
Drag Coefficient（阻力系数）：气动阻力用单位面积因数、载具速度和恒定空气密度计算。该值乘到这一计算结果上。力的方向始终与载具速度相反。

• DownForce Coefficient : Just like drag force, the calculated down force is multiplied with this value. The direction of the down force is against the gravity.
DownForce Coefficient（下压力系数）：与阻力一样，计算出的下压力会乘以该值。下压力的方向与重力相反。

• DownForce X-Offset : If you want the down force to be applied more to the front of the vehicle give a positive value in meters and negative for the rear side.
DownForce X-Offset（下压力 X 偏移）：如果希望下压力更多地施加在载具前部，给出正值，单位为米；后部则用负值。

StandardInput Settings

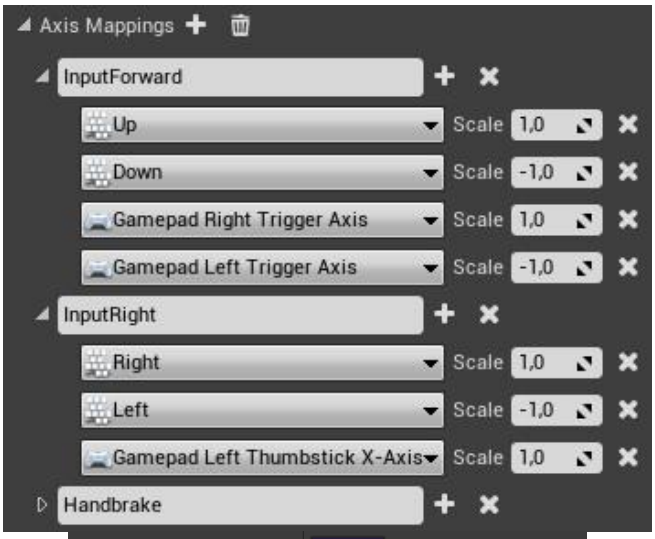
标准输入设置

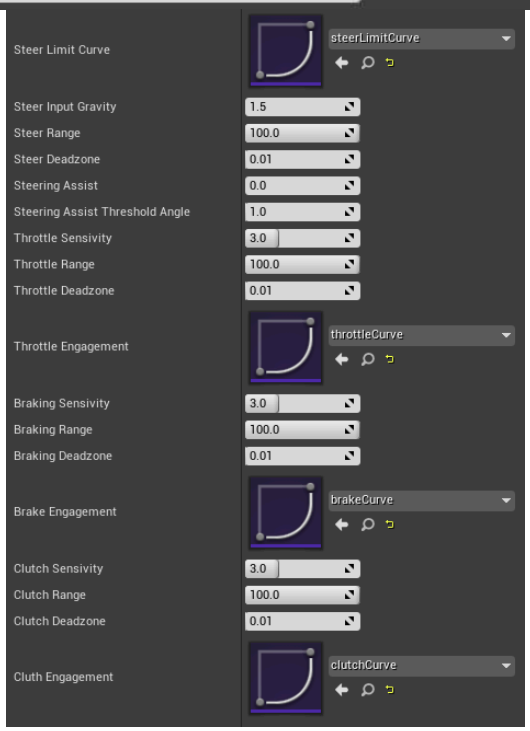
To use the standard input component(with the legacy input system) first you need to define axis mappings in your project settings. Below image shows a sample setup for some of the inputs. Steering wheel input is tested with a logitech G29 wheel and if you want to use a steering wheel you need to activate the raw input plugin* of unreal. After setting up the axis mappings the rest is pretty easy.

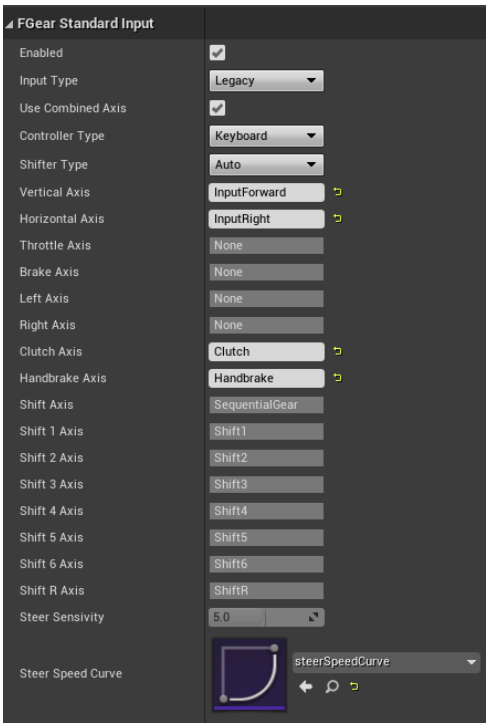
要使用标准输入组件（配合传统输入系统），首先需要在项目设置中定义轴映射。下图是部分输入的示例设置。方向盘输入用罗技 G29 测试过；若要使用方向盘，需要启用虚幻的原始输入插件*。轴映射设好之后，其余就很直接。

*https://docs.unrealengine.com/en-US/Gameplay/Input/RawInput/index.html

*https://docs.unrealengine.com/en-US/Gameplay/Input/RawInput/index.html







• Enabled : If unchecked user inputs will not have any effect. For example if you want your vehicle to be controlled by AI then you should uncheck this.
Enabled（启用）：若不勾选，用户输入不会有任何效果。例如希望载具由 AI 控制时，应取消勾选。

• Input Type : Whether to use legacy input system or the enhanced input system. See "Enhanced Input System" section below.
Input Type（输入类型）：使用传统输入系统还是增强输入系统。见下方 "Enhanced Input System" 一节。

• Use Combined Axis : This applies only to throttle, brake and steering controls. If you want to use a combined axis like Vertical Axis for throttle/brake controls and Horizontal Axis for steering then enable this. If you want to use seperate axis for those then uncheck this option and setup proper axis values.
Use Combined Axis（使用合并轴）：只作用于油门、制动和转向。如果要用 Vertical Axis 这类合并轴做油门/制动，用 Horizontal Axis 做转向，则启用。如果要为它们使用分开的轴，则取消该选项并设置对应的轴。

• Controller : Type of the controller : Keyboard, joystick or wheel. Most of the controls can work at the same time, for ex. even if you select keyboard you can play with joystick too.
Controller（控制器）：控制器类型：键盘、手柄或方向盘。大多数控制可以同时工作，例如即使选择键盘，也可以同时用手柄玩。

• Shifter Type : Type of the shifter : Auto, sequential or manual. If the AutoChange option of the transmission is checked this selection has no effect but if you want to manually change gears then you can choose a sequential mode or full manual mode.
Shifter Type（换挡器类型）：换挡器类型：自动、序列式或手动。若变速箱的 Auto Change 已勾选，该选择没有效果；若要手动换挡，可以选择序列式或完全手动。

• Vertical Axis : This is a combined axis for throttle and brakes. Range : [-1, 1].
Vertical Axis（垂直轴）：油门和制动的合并轴。范围：[-1, 1]。

• Horizontal Axis : This is the combined steering axis name. Range : [-1, 1].
Horizontal Axis（水平轴）：合并转向轴的名称。范围：[-1, 1]。

• Throttle Axis : If you uncheck Use Combined Axis option you can assign an axis for throttle with this. Range : [0, 1].
Throttle Axis（油门轴）：若取消 Use Combined Axis，可以用该项指定油门轴。范围：[0, 1]。

• Brake Axis : If you uncheck Use Combined Axis option you can assign an axis for braking with this. Range :

[0, 1].
Brake Axis（制动轴）：若取消 Use Combined Axis，可以用该项指定制动轴。范围：[0, 1]。

• Left Axis : If you uncheck Use Combined Axis option you can assign an axis for left steering with this. Range : [0, 1].
Left Axis（左转向轴）：若取消 Use Combined Axis，可以用该项指定左转向轴。范围：[0, 1]。

• Right Axis : If you uncheck Use Combined Axis option you can assign an axis for right steering with this. Range : [0, 1].
Right Axis（右转向轴）：若取消 Use Combined Axis，可以用该项指定右转向轴。范围：[0, 1]。

• Clutch Axis : Clutch axis for manual transmission. Range : [-1, 1] but internally normalized to [0,1].
Clutch Axis（离合器轴）：手动变速箱的离合器轴。范围：[-1, 1]，但内部会归一化到 [0, 1]。

• Handbrake Axis : Handbrake input axis. Range : [0, 1].
Handbrake Axis（手刹轴）：手刹输入轴。范围：[0, 1]。

• Shift Axis : This is a combined axis for gear up and gear down. Range : [-1, 1]. This is only activated in sequential gear mode.
Shift Axis（换挡轴）：升挡和降挡的合并轴。范围：[-1, 1]。仅在序列式挡位模式下启用。

• Shift 1/2/../R Axis : Axis names for each gear. Range : [0, 1]. These are only activated in manual gear mode.
Shift 1/2/../R Axis（各挡轴）：每个挡位的轴名称。范围：[0, 1]。仅在手动挡位模式下启用。

• Steer Speed Curve : If you want the steering speed to decrease as the vehicle goes faster you can use this curve. The x-axis represents speed in kmh and the y-axis is the steering speed.
Steer Speed Curve（转向速度曲线）：如果希望车速越快转向速度越低，可以使用该曲线。X 轴是速度（km/h），Y 轴是转向速度。

• Steer Limit Curve : If you want the maximum steering angle to be limited as the vehicle goes faster you can use this curve. The x-axis represents speed in kmh and the y-axis should be a value between 0 and 100. 100 means steering is not limited and 0 means no steering possible at all. The limit angle is calculated as (value/100) * MaxSteerAngle.
Steer Limit Curve（转向限制曲线）：如果希望车速越快最大转向角越受限制，可以使用该曲线。X 轴是速度（km/h），Y 轴应在 0 到 100 之间。100 表示转向不受限制，0 表示完全不能转向。限制角按 (值/100) × MaxSteerAngle 计算。

• Steer Input Gravity : If the current steering angle and the desired steering angle have different signs than current steering speed is multiplied with this value. This lets user to counter steer more easily with the keyboard or joystick.
Steer Input Gravity（转向输入回中）：若当前转向角与期望转向角符号不同，当前转向速度会乘以该值。这让键盘或手柄更容易做反打。

• Steer Sensivity : This value is only used when the controller is a wheel. This determines how fast the steering value catches up with the input axis value. Higher values result sharper steering.
Steer Sensivity（转向灵敏度）：仅当控制器是方向盘时使用。它决定转向值多快跟上输入轴的值。数值越高转向越锐利。

• Steer Range : The percentage of steering range. For ex. if you set this to %10 then a %10 input will result %100 steering. Another example is that a steering wheel with 900 degress of working range needs 450 degrees of rotation to make a full turn but if this value is %10 then a 45 degrees of rotation will make a full turn.
Steer Range（转向范围）：转向范围的百分比。例如设为 10%，则 10% 的输入会产生 100% 的转向。另一个例子是，工作范围 900 度的方向盘需要转 450 度才能打满；若该值为 10%，则转 45 度就会打满。

• Steer Deadzone : Input values below this value are discarded. A small dead zone is mostly usefull.
Steer Deadzone（转向死区）：低于该值的输入被丢弃。一个小死区通常有用。

• Steering Assist : Steering assist tries to steer towards the direction that the vehicle body travels.

Steering Assist（转向辅助）：转向辅助会尝试朝车身实际运动的方向转向。

• Steering Assist Threshold : If the angle between the steering direction and the vehicles velocity direction is above this value, steering assist is activated.
Steering Assist Threshold（转向辅助阈值）：若转向方向与载具速度方向之间的夹角高于该值，则启用转向辅助。

• Throttle Sensivity : This determines how fast the throttle value catches up with the input axis value. Higher values result sharper throttle.
Throttle Sensivity（油门灵敏度）：决定油门值多快跟上输入轴的值。数值越高油门越锐利。

• Throttle Range : The percentage of throttle range. Works similar to steering range.
Throttle Range（油门范围）：油门范围的百分比。作用与转向范围类似。

• Throttle Deadzone : Input values below this value are discarded. A small dead zone is mostly usefull.
Throttle Deadzone（油门死区）：低于该值的输入被丢弃。一个小死区通常有用。

• Throttle Engagement : Throttle input takes values between 0 and 1. If throttle is fully pressed it is 1 and if it is not pressed at all then it is 0. This value changes linearly as you press the pedal. The throttle input represents the X-Axis of this curve and the resulting value represents the real engine throttle. Throttle pedal behaviour changes from vehicle to vehicle so you can customise it with this curve. The default curve is a linear 1 to 1 pedal.
Throttle Engagement（油门接合曲线）：油门输入取值在 0 到 1 之间。完全踩下为 1，完全不踩为 0。踩踏板时该值线性变化。油门输入是该曲线的 X 轴，结果值是真正的发动机油门。不同载具的油门踏板特性不同，可以用该曲线定制。默认曲线是 1 对 1 的线性踏板。

• Braking Sensivity : This determines how fast the braking value catches up with the input axis value. Higher values result sharper braking.
Braking Sensivity（制动灵敏度）：决定制动值多快跟上输入轴的值。数值越高制动越锐利。

• Braking Range : The percentage of braking range. Works similar to steering range.
Braking Range（制动范围）：制动范围的百分比。作用与转向范围类似。

• Braking Deadzone : Input values below this value are discarded. A small dead zone is mostly usefull.
Braking Deadzone（制动死区）：低于该值的输入被丢弃。一个小死区通常有用。

• Brake Engagement : Brake input takes values between 0 and 1. If brake is fully pressed it is 1 and if it is not pressed at all then it is 0. This value changes linearly as you press the pedal. The brake input represents the X-Axis of this curve and the resulting value represents the real brake power ratio. Brake pedal behaviour changes from vehicle to vehicle so you can customise it with this curve. The default curve is a linear pedal.
Brake Engagement（制动接合曲线）：制动输入取值在 0 到 1 之间。完全踩下为 1，完全不踩为 0。踩踏板时该值线性变化。制动输入是该曲线的 X 轴，结果值是真正的制动力比例。不同载具的制动踏板特性不同，可以用该曲线定制。默认曲线是线性踏板。

• Clutch Sensivity : This determines how fast the clutch value catches up with the input axis value. Higher values result sharper clutch engagement.
Clutch Sensivity（离合器灵敏度）：决定离合器值多快跟上输入轴的值。数值越高离合器接合越锐利。

• Clutch Range : The percentage of clutch range. Works similar to steering range.
Clutch Range（离合器范围）：离合器范围的百分比。作用与转向范围类似。

• Clutch Deadzone : Input values below this value are discarded. A small dead zone is mostly usefull.
Clutch Deadzone（离合器死区）：低于该值的输入被丢弃。一个小死区通常有用。

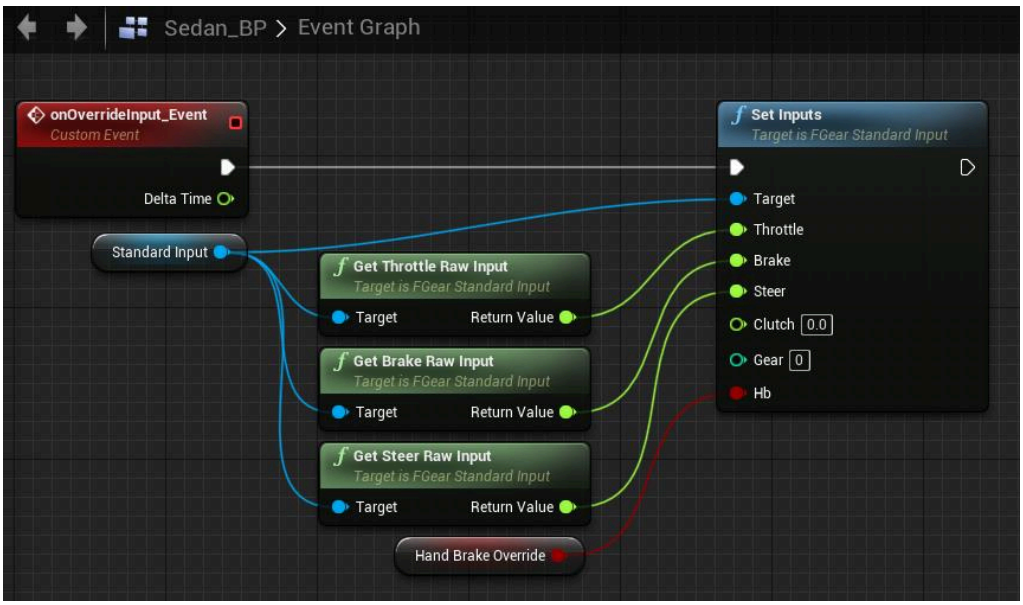
• Clutch Engagement : There is an internal variable called ClutchState which takes values between 0 and 1. If clutch is full pressed it is 1 and if it is not pressed at all then it is 0. This value changes linearly as you press the pedal. The ClutchState represents the X-Axis of this curve and the resulting value represent the real clutch power ratio. Clutch pedal behaviour changes from vehicle to vehicle but generally try to leave a dead zone before engaging and then quickly raise the power ratio. If you do not provide a curve a default curve is

created which is a simple example to this.

Clutch Engagement（离合器接合曲线）：有一个名为 ClutchState 的内部变量，取值在 0 到 1 之间。离合器完全踩下为 1，完全不踩为 0。踩踏板时该值线性变化。ClutchState 是该曲线的 X 轴，结果值是真正的离合器动力比例。不同载具的离合器踏板特性不同，但通常应在接合前留一段死区，然后迅速提高动力比例。若不提供曲线，会创建一条默认曲线作为简单示例。

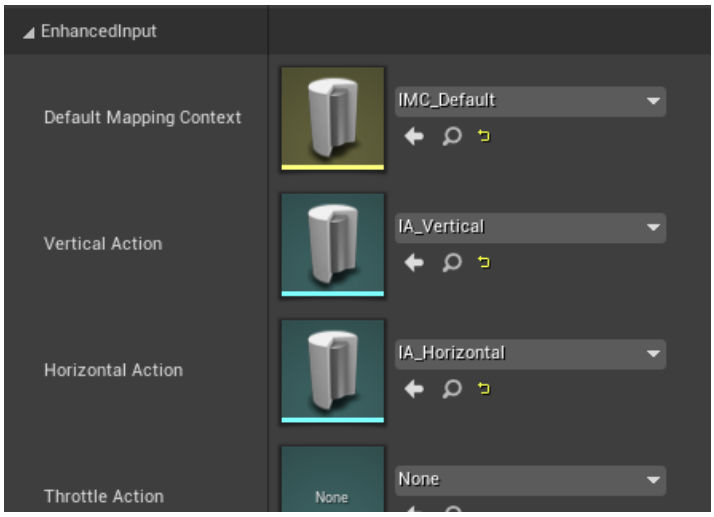
Input Overriding : If you want to override standard input results without disabling you can listen to the "onOverrideInput" event of the vehicle and make changes you need. Check "VehicleBP/Sedan_BP" or "UI/MobileUI" for an example. In the image below we override the handbrake input and we keep the original inputs for throttle, brake and steering.

Input Overriding（输入覆盖）：如果想在不禁用标准输入的情况下覆盖其结果，可以监听载具的 "onOverrideInput" 事件并做所需修改。示例见 "VehicleBP/Sedan_BP" 或 "UI/MobileUI"。下图中覆盖了手刹输入，并保留油门、制动和转向的原始输入。



Enhanced Input System : Standard input component supports the enhanced input system. First you need to activate enhanced input system for your project. Go to the projects settings and locate input section and find the default classes settings. Change the input classes to enhanced input versions. Then set "InputType" option to "Enhanced". Enhanced input requires an input mapping context and a set of input actions. There is an input action slot for each corresponding axis. The sample project includes some basic assets for demonstration under "FGearAssets/Inputs" folder. When you have the necessary assets you can set them in the standard input settings like below.

Enhanced Input System（增强输入系统）：标准输入组件支持增强输入系统。首先要为工程启用增强输入。打开项目设置，找到 Input 一节中的默认类设置，把输入类改成增强输入版本。然后把 "InputType" 设为 "Enhanced"。增强输入需要一个输入映射情境和一组输入动作。每个对应的轴都有一个输入动作槽。示例工程在 "FGearAssets/Inputs" 文件夹中包含一些用于演示的基础资产。准备好资产后，可以像下面这样在标准输入设置中指定它们。



Replication Settings

复制设置

FGear replication has two main tasks:

FGear 的复制有两项主要任务：

1. Input replication :

1. 输入复制：

a. ServerAuthoritative: Local players send inputs to the server, server replicates back. b. ClientAuthoritative: Local players send inputs to every other node.

a. ServerAuthoritative（服务器权威）：本地玩家把输入发给服务器，服务器再复制回来。b. ClientAuthoritative（客户端权威）：本地玩家把输入发给其他每一个节点。

2. State replication :

2. 状态复制：

a. ServerAuthoritative: Server sends kinematic states to the clients, clients use a dead reckoning algorithm to make things smooth. b. ClientAuthoritative: Each client simulates locally and sends kinematic states to the others, others use a dead reckoning algorithm to make things smooth.

a. ServerAuthoritative（服务器权威）：服务器把运动学状态发给客户端，客户端用航位推算使表现平滑。b. ClientAuthoritative（客户端权威）：每个客户端在本地模拟，并把运动学状态发给其他端，其他端用航位推算使表现平滑。

Warning : Server authoritative replication has no client prediction yet.

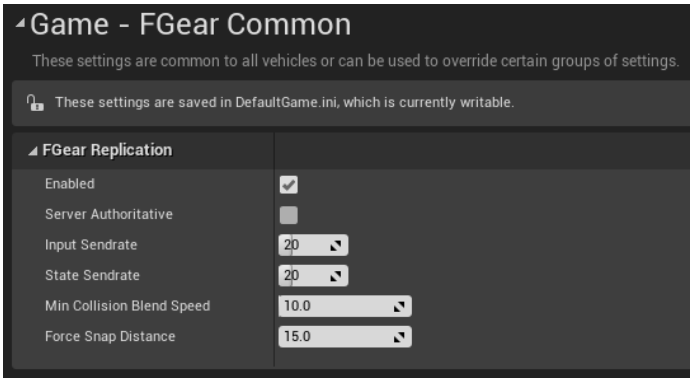
警告：服务器权威复制目前还没有客户端预测。

Every vehicle share these common settings unless they’re set to override.

除非载具被设为覆盖，否则每辆载具都共用这些设置。

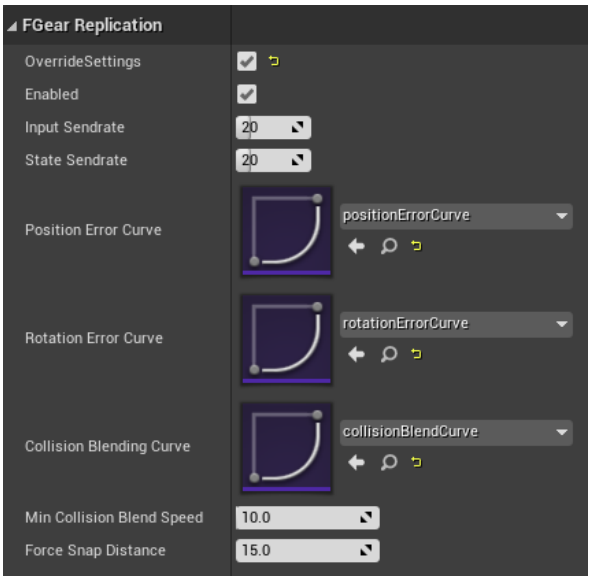
ProjectSettings->Game->FGearCommon

Project Settings → Game → FGearCommon



It is possible to override the common settings on each vehicle.
可以在每辆载具上覆盖这些共用设置。

Note : We create default curves in code if the user do not provide any curve. The default curves work well for most cases.
注意：若用户没有提供曲线，我们会在代码中创建默认曲线。默认曲线在多数情况下表现良好。



• OverrideSettings : If set you can skip the common settings and use your desired settings for this vehicle.
OverrideSettings（覆盖设置）：若设置，可以跳过共用设置，对该载具使用自己想要的设置。

• Enabled : If unchecked fgear replication has no effect at all.
Enabled（启用）：若不勾选，FGear 复制完全不起作用。

• Server Authoritative : By default the client authoritative system is used. If this is set then the legacy server authoritative system is used instead.
Server Authoritative（服务器权威）：默认使用客户端权威系统。若设置该项，则改用旧的服务器权威系统。

• Input Sendrate : Send inputs at this rate, for ex. 20 per second means an rpc is called every 50 miliseconds.
Input Sendrate（输入发送频率）：按该频率发送输入。例如每秒 20 次表示每 50 毫秒调用一次 RPC。

• State Sendrate : Send states at this rate, for ex. 20 per second means an rpc is called every 50 miliseconds. The dead reckoning algorithm works fine with lower values so you can keep this lower than input sendrate for optimization.
State Sendrate（状态发送频率）：按该频率发送状态。例如每秒 20 次表示每 50 毫秒调用一次 RPC。航位推算在更低的值下也能工作，因此为了优化可以让它低于输入发送频率。

• Position Error Curve : This parameter is only valid for the client authoritative mode. Client authoritative replication has the downside of issues with collision. In order to handle the collisions more smoothly we try to blend the network transform with the physics transform. This blend is affected by multiple factors. The first one is the position error. This error is the distance between the physics location and the network location so the error gets larger as the physics location gets further away from the network location. This curve tells us how much we blend with physics depending on the position error. The X axis of the curve represents the position error in meters and the Y axis of the curve is the blend weight which should be between 0 and 1. When you do not provide a curve the system creates a default curve for you which works well for most cases. This is how the default curve is set:
Position Error Curve（位置误差曲线）：该参数仅对客户端权威模式有效。客户端权威复制的缺点是碰撞问题。为了更平滑地处理碰撞，我们尝试把网络变换与物理变换混合。该混合受多个因素影响。第一个是位置误差，即物理位置与网络位置之间的距离，物理位置离网络位置越远，误差越大。该曲线根据位置误差说明与物理混合多少。曲线 X 轴是位置误差（米），Y 轴是混合权重，应在 0 到 1 之间。若不提供曲线，系统会创建一条在多数情况下表现良好的默认曲线。默认曲线这样设置：

• At zero error the blend weight is also zero which means full physics blending.
误差为零时，混合权重也是零，表示完全使用物理混合。

• At 0.5 meters of error the blend weight is 0.1 so we still have %90 physics weight.
误差为 0.5 米时，混合权重为 0.1，因此仍有 90% 的物理权重。

• At 1.0 meters and above the blend weight is 1.0 which means zero physics weight.
误差为 1.0 米及以上时，混合权重为 1.0，表示物理权重为零。

So it is better to let the physics drive the transform for small errors but we need to quickly enforce the network transform as the error gets larger. There is also a sample curve in the example project: “FGearAssets/Curves/positionErrorCurve”.
因此小误差时最好让物理驱动变换，但误差变大时需要迅速强制使用网络变换。示例工程中还有一条示例曲线：“FGearAssets/Curves/positionErrorCurve”。

• Rotation Error Curve : This parameter is only valid for the client authoritative mode. Rotation error is the angular distance in degrees between the physics rotation and the network rotation. This curve tells us how much we blend with physics depending on the rotation error. The X axis of the curve represents the rotation error in degrees and the Y axis of the curve is the blend weight which should be between 0 and 1. When you do not provide a curve the system creates a default curve for you which works well for most cases. This is how the default curve is set:
Rotation Error Curve（旋转误差曲线）：该参数仅对客户端权威模式有效。旋转误差是物理旋转与网络旋转之间的角距离，单位为度。该曲线根据旋转误差说明与物理混合多少。曲线 X 轴是旋转误差（度），Y 轴是混合权重，应在 0 到 1 之间。若不提供曲线，系统会创建一条在多数情况下表现良好的默认曲线。默认曲线这样设置：

• At zero error the blend weight is also zero which means full physics blending.
误差为零时，混合权重也是零，表示完全使用物理混合。

• At 5 degrees of error the blend weight is 0.1 so we still have %90 physics weight.
误差为 5 度时，混合权重为 0.1，因此仍有 90% 的物理权重。

• At 15 degrees and above the blend weight is 1.0 which means zero physics weight.
误差为 15 度及以上时，混合权重为 1.0，表示物理权重为零。

There is also a sample curve in the project: “FGearAssets/Curves/rotationErrorCurve”.
工程中还有一条示例曲线：“FGearAssets/Curves/rotationErrorCurve”。

• Collision Blending Curve : This parameter is only valid for the client authoritative mode. This curve defines an overall factor for physics blending that scales both position and rotation error weights. The idea is that when a collision occurs we can trigger a full physics blend and then it fades out back to network transform. This approach helps to get better collision responses. So this curve defines the duration and the weight of the overall physics blending scale. The X axis of the curve represents the time in seconds and the Y axis of the curve is the scale of both position & rotation errors which should be between 0 and 1. When you do not provide a curve the system creates a default curve for you which works well for most cases. This is how the default curve is set:
Collision Blending Curve（碰撞混合曲线）：该参数仅对客户端权威模式有效。该曲线定义物理混合的总系数，同时缩放位置误差权重和旋转误差权重。思路是碰撞发生时可以触发一次完全的物理混合，然后再淡出回到网络变换。这有助于得到更好的碰撞响应。因此该曲线定义整体物理混合缩放的持续时间和权重。曲线 X 轴是时间（秒），Y 轴是位置与旋转误差的缩放，应在 0 到 1 之间。若不提供曲线，系统会创建一条在多数情况下表现良好的默认曲线。默认曲线这样设置：

• At zero time we have zero error scale which means %100 physics blend.
时间为零时，误差缩放为零，表示 100% 物理混合。

• At 0.5 seconds we have 0.1 error scale which is still %90 error scale.
0.5 秒时，误差缩放为 0.1，因此仍是 90% 的误差缩放。

• At 1.0 seconds we have 1.0 error scale which is back to %0 physics blend.
1.0 秒时，误差缩放为 1.0，回到 0% 物理混合。

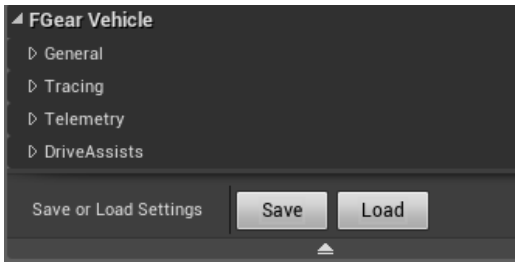
So we start with full physics blend and slowly fade out for a short time and then quickly recover back to full network transform. There is also a sample curve in the project: “FGearAssets/Curves/collisionBlendCurve”.
因此我们从完全的物理混合开始，短时间内缓慢淡出，然后迅速恢复到完全的网络变换。工程中还有一条示例曲线：“FGearAssets/Curves/collisionBlendCurve”。

• **Min Collision Blend Speed : This parameter is only valid for the client authoritative mode. Collision blending is only triggered if the relative speed at the contact point is greater then this value(in kmh). If you use a small number then every contact may result a full physics blend which is not ideal and a large number may prevent a required blend.**
Min Collision Blend Speed（最小碰撞混合速度）：该参数仅对客户端权威模式有效。仅当接触点的相对速度大于该值（km/h）时才触发碰撞混合。数值太小会使每次接触都变成完全的物理混合，这不理想；数值太大则可能阻止必要的混合。

• **Force Snap Distance : This parameter is only valid for the client authoritative mode. When the network position error is larger then this value(in meters) then we skip the local collision check(sweep) and force a snap(teleport).**
Force Snap Distance（强制吸附距离）：该参数仅对客户端权威模式有效。当网络位置误差大于该值（米）时，跳过本地碰撞检测（扫描），并强制吸附（传送）。

WARNING : In order to trigger physics blending with collisions we should be able to receive hit events within the vehicle. If the collision profile of the vehicle is properly set(both colliders should block and check “Simulation Generates Hit Events”) then we should be fine but if that’s not the case the blending will never happen. It might be difficult to distinguish the difference especially when the network conditions are good so we added a new telemetry option called “PrintCollisionLogs” in order to make sure you’re getting those hit events. When this option is on, the valid collision events will print a log on the console like “NotifyHit Vehicle1<->Vehicle2….”. If you are not getting any logs then it means the physics blending is not happening so you should check your collision profiling setup and also make sure the “Min Collision Blend Speed” condition is met for the contacts.
警告：为了在碰撞时触发物理混合，载具内部必须能收到命中事件。若载具的碰撞预设设置正确（两个碰撞体都应阻挡，并勾选 “Simulation Generates Hit Events”），就应当没有问题；否则混合永远不会发生。网络状况良好时尤其难以分辨差别，因此增加了名为 “PrintCollisionLogs” 的遥测选项，用来确认正在收到这些命中事件。启用后，有效的碰撞事件会在控制台打印类似 “NotifyHit Vehicle1<->Vehicle2….” 的日志。如果没有任何日志，就说明物理混合没有发生，应检查碰撞预设，并确认接触满足 “Min Collision Blend Speed” 条件。

Save/Load Vehicles
保存/加载载具



Warning : This is only intended to be used as an editor tool not to be used in realtime. Additionally due to a problem with the unreal editor, it can be problematic to save/load from the blueprint editor window. When blueprint window is open and the viewport tab is not selected, extra axle components(axles other then front/rear) can not be retrieved so it is safer to save/load from regular details panel or make sure the viewport panel is active in blueprint window. Also when you load a configuration the changes will not effect the instances in a scene so you need to click the “Reset Instance Changes to Blueprint Default” button.
警告：这只打算作为编辑器工具，不要用于实时运行。另外，由于虚幻编辑器的一个问题，从蓝图编辑器窗口保存/加载可能有问题。蓝图窗口打开且没有选中视口页签时，取不到额外的车轴组件（前轴/后轴以

外的车轴），因此更安全的做法是从普通细节面板保存/加载，或确保蓝图窗口中的视口面板处于激活状态。加载配置后，改动也不会影响场景中的实例，因此需要点击 “Reset Instance Changes to Blueprint Default” 按钮。

- **Save : Asks for a save location and creates a json file that contains all serialized fields of the vehicle.**
  Save（保存）：询问保存位置，并创建一个包含该载具全部已序列化字段的 JSON 文件。
- **Load : Loads the selected json file, retrieves the saved settings and applies to the current vehicle.**
  Load（加载）：加载所选 JSON 文件，取出保存的设置并应用到当前载具。

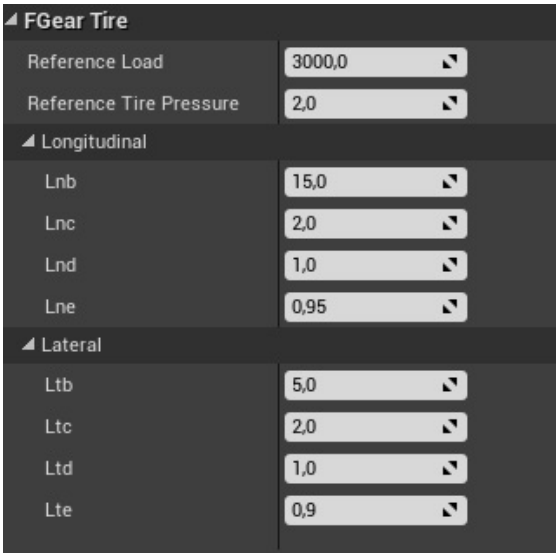
Tire Models

轮胎模型

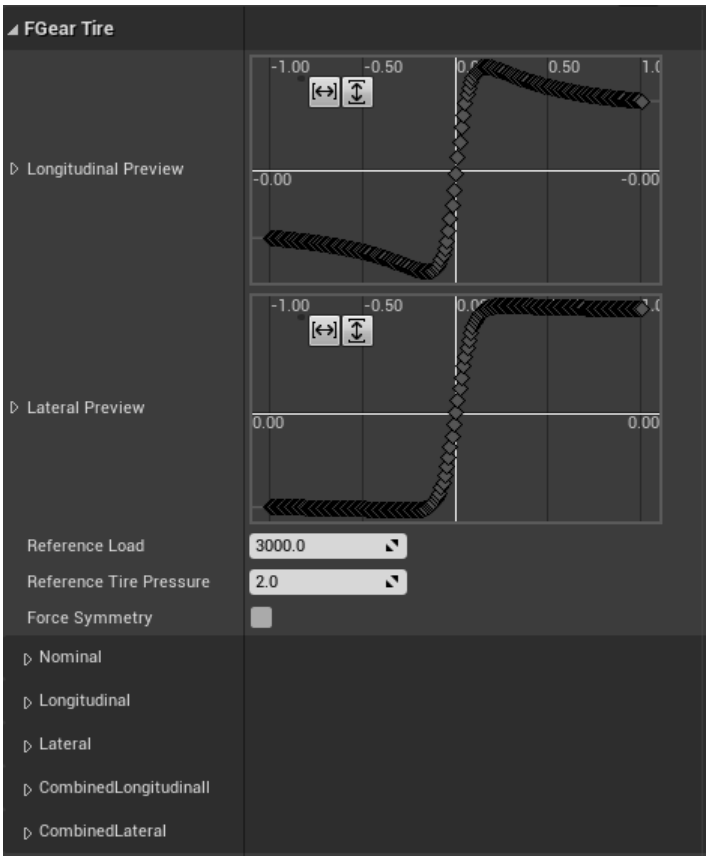
FGear uses pacejka’s magic formula tire modelling approach. This is a curve fitting algorithm and it has no releation to physical properties of a tire but it has great precision and widely used in lots of fields. Pacejka’s formula has multiple versions but the fundamentals are the same. FGear uses pacejka96 and MF6.1 version but a simplified version is also available. The default tire model classes (FGearTireSimple, FGearTire96, FGearTireMF61) can be used right away. If you want to make a new tire model you can simply create a new blueprint class and derive from one of the tire mode classes. All available tire models will be listed in wheel options.
FGear 使用 Pacejka 魔术公式的轮胎建模方法。这是一种曲线拟合算法，与轮胎的物理属性没有关系，但精度很高，并在很多领域广泛使用。Pacejka 公式有多个版本，但基本原理相同。FGear 使用 Pacejka 96 和 MF6.1，同时也提供一个简化版本。默认轮胎模型类（FGearTireSimple、FGearTire96、FGearTireMF61）可以直接使用。若要制作新的轮胎模型，只需新建蓝图类并从其中一个轮胎模型类派生。所有可用的轮胎模型都会列在车轮选项中。

TireSimple : This version has only 4 parameters(B,C,D,E) and is independent from load or camber.
TireSimple：该版本只有 4 个参数（B、C、D、E），并且与载荷或外倾无关。

B : Stiffness - C : Shape - D : Peak - E : Curvature
B：刚度。C：形状。D：峰值。E：曲率。



Tire96 : This is the pacejka96 tire model and it has many parameters. There are lots resources available on the internet about tire models so if you want to learn its details you can easily find them but the real deal is about finding good parameter sets. Real pacejka sets are not free and not easy to come by. The default parameters provided in this package are a good starting point for you to tweak and also check the link below for a couple of lateral parameter sets : http://www.avonmotorsport.com/resource-centre/downloads
Tire96：这是 Pacejka 96 轮胎模型，参数很多。网上有大量关于轮胎模型的资料，想了解细节很容易找到，但真正困难的是找到好的参数组。真实的 Pacejka 参数组并不免费，也不容易得到。本包提供的默认参数是一个很好的调整起点。下面的链接还有几组侧向参数：http://www.avonmotorsport.com/resource-centre/downloads



List of longitudinal friction pure slip paremeters:
纵向摩擦纯滑移参数列表:

PCX1 : Shape factor Cfx for longitudinal force PDX1 : Longitudinal friction Mux at Fznom PDX2 : Variation of friction Mux with load PEX1 : Longitudinal curvature Efx at Fznom PEX2 : Variation of curvature Efx with load PEX3 : Variation of curvature Efx with load squared PEX4 : Factor in curvature Efx while driving PKX1 : Longitudinal slip stiffness Kfx/Fz at Fznom PKX2 : Variation of slip stiffness Kfx/Fz with load PKX3 : Exponent in slip stiffness Kfx/Fz with load PHX1 : Horizontal shift Shx at Fznom PHX2 : Variation of shift Shx with load PVX1 : Vertical shift Svx/Fz at Fznom PVX2 : Variation of shift Svx/Fz with load
PCX1：纵向力的形状系数 Cfx。PDX1：名义载荷 Fznom 下的纵向摩擦 Mux。PDX2：摩擦 Mux 随载荷的变化。PEX1：名义载荷 Fznom 下的纵向曲率 Efx。PEX2：曲率 Efx 随载荷的变化。PEX3：曲率 Efx 随载荷平方的变化。PEX4：驱动时曲率 Efx 中的系数。PKX1：名义载荷 Fznom 下的纵向滑移刚度 Kfx/Fz。PKX2：滑移刚度 Kfx/Fz 随载荷的变化。PKX3：滑移刚度 Kfx/Fz 随载荷变化的指数。PHX1：名义载荷 Fznom 下的水平偏移 Shx。PHX2：偏移 Shx 随载荷的变化。PVX1：名义载荷 Fznom 下的垂直偏移 Svx/Fz。PVX2：偏移 Svx/Fz 随载荷的变化。

List of lateral friction pure slip paremeters:
侧向摩擦纯滑移参数列表:

PCY1 : Shape factor Cfy for lateral forces PDY1 : Lateral friction Muy PDY2 : Variation of friction Muy with load PDY3 : Variation of friction Muy with squared inclination PEY1 : Lateral curvature Efy at Fznom PEY2 : Variation of curvature Efy with load PEY3 : Inclination dependency of curvature Efy PEY4 : Variation of curvature Efy with inclination PKY1 : Maximum value of stiffness Kfy/Fznom PKY2 : Load at which Kfy reaches maximum value PKY3 : Variation of Kfy/Fznom with inclination PHY1 : Horizontal shift Shy at Fznom PHY2 : Variation of shift Shy with load PHY3 : Variation of shift Shy with inclination PVY1 : Vertical shift in Svy/Fz at Fznom PVY2 : Variation of shift Svy/Fz with load PVY3 : Variation of shift Svy/Fz with inclination PVY4 : Variation of shift Svy/Fz with inclination and load
PCY1：侧向力的形状系数 Cfy。PDY1：侧向摩擦 Muy。PDY2：摩擦 Muy 随载荷的变化。PDY3：摩擦 Muy 随倾角平方的变化。PEY1：名义载荷 Fznom 下的侧向曲率 Efy。PEY2：曲率 Efy 随载荷的变化。PEY3：曲率 Efy 对倾角的依赖。PEY4：曲率 Efy 随倾角的变化。PKY1：刚度 Kfy/Fznom 的最大值。PKY2：Kfy 达到最大值时的载荷。PKY3：Kfy/Fznom 随倾角的变化。PHY1：名义载荷 Fznom 下的水平偏移 Shy。PHY2：偏移 Shy 随载荷的变化。PHY3：偏移 Shy 随倾角的变化。PVY1：名义载荷 Fznom 下 Svy/Fz 的垂直偏移。PVY2：偏移 Svy/Fz 随载荷的变化。PVY3：偏移 Svy/Fz 随倾角的变化。PVY4：偏移 Svy/Fz 随倾角和载荷的

变化。

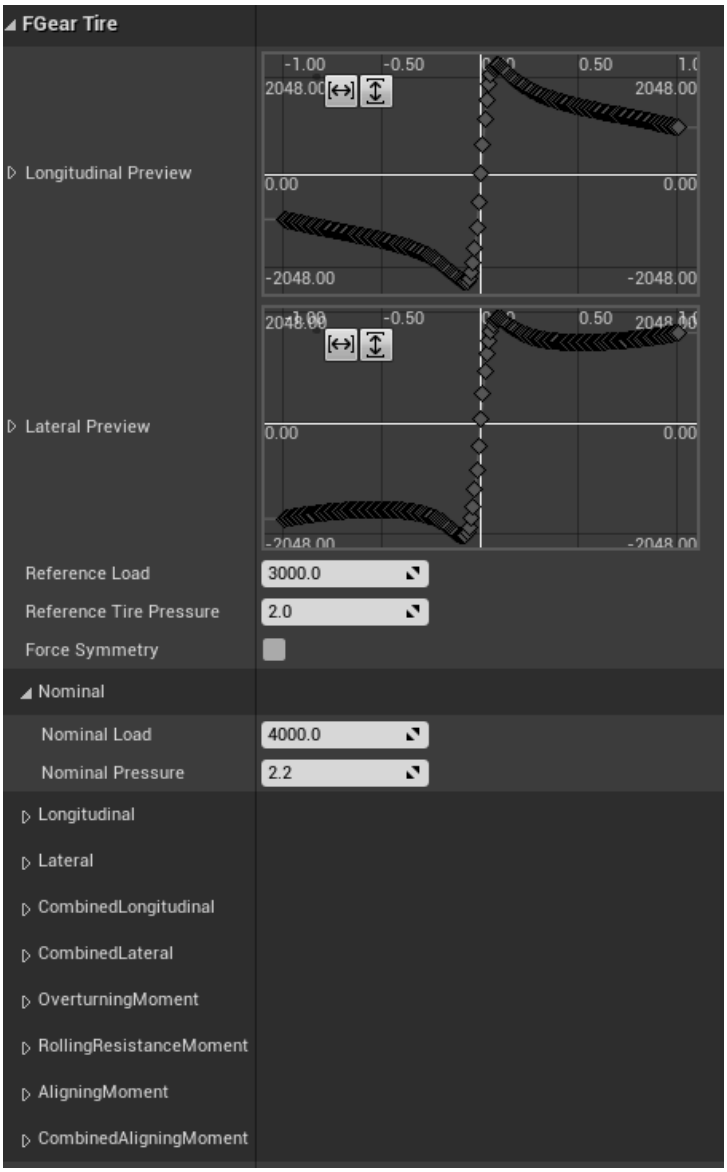
List of longitudinal friction combined slip paremeters:
纵向摩擦联合滑移参数列表:

RBX1 : Slope factor for combined slip Fx reduction RBX2 : Variation of slope Fx reduction with kappa RCX1 : Shape factor for combined slip Fx reduction REX1 : Curvature factor of combined Fx REX2 : Curvature factor of combined Fx with load RHX1 : Shift factor for combined slip Fx reduction
RBX1：联合滑移 Fx 衰减的斜率系数。RBX2：Fx 衰减斜率随 kappa 的变化。RCX1：联合滑移 Fx 衰减的形状系数。REX1：联合 Fx 的曲率系数。REX2：联合 Fx 曲率系数随载荷的变化。RHX1：联合滑移 Fx 衰减的偏移系数。

List of lateral friction combined slip paremeters:
侧向摩擦联合滑移参数列表:

RBY1 : Slope factor for combined Fy reduction RBY2 : Variation of slope Fy reduction with alpha RBY3 : Shift term for alpha in slope Fy reduction RCY1 : Shape factor for combined Fy reduction REY1 : Curvature factor of combined Fy REY2 : Curvature factor of combined Fy with load RHY1 : Shift factor for combined Fy reduction RVY1 : Kappa induced side force Svyk/Muy*Fz at Fznom RVY2 : Variation of Svyk/Muy*Fz with load RVY3 : Variation of Svyk/Muy*Fz with inclination RVY4 : Variation of Svyk/Muy*Fz with alpha RVY5 : Variation of Svyk/Muy*Fz with kappa RVY6 : Variation of Svyk/Muy*Fz with atan (kappa)
RBY1：联合 Fy 衰减的斜率系数。RBY2：Fy 衰减斜率随 alpha 的变化。RBY3：Fy 衰减斜率中 alpha 的偏移项。RCY1：联合 Fy 衰减的形状系数。REY1：联合 Fy 的曲率系数。REY2：联合 Fy 曲率系数随载荷的变化。RHY1：联合 Fy 衰减的偏移系数。RVY1：名义载荷 Fznom 下由 kappa 引起的侧向力 Svyk/Muy*Fz。RVY2：Svyk/Muy*Fz 随载荷的变化。RVY3：Svyk/Muy*Fz 随倾角的变化。RVY4：Svyk/Muy*Fz 随 alpha 的变化。RVY5：Svyk/Muy*Fz 随 kappa 的变化。RVY6：Svyk/Muy*Fz 随 atan(kappa) 的变化。

TireMF61 :This is the magic formula 6.1 tire model and it has many more parameters(over 100). Tire96 is limited to Fx and Fy outputs but this one also generates momentum effects namely Mx, My and Mz. The details of this model is covered in Pacejkas Tire and Vehicle Dynamics book. The default parameter set is taken from the book but the parameters of pacejka 96 could also be adapted. The reference parameters like Reference Load, Reference Camber and Reference Tire Pressure are for previewing so they do not effect the tire model and they are not saved.
TireMF61：这是魔术公式 6.1 轮胎模型，参数更多（超过 100 个）。Tire96 只输出 Fx 和 Fy，而该模型还会产生力矩效果，即 Mx、My 和 Mz。该模型的细节写在 Pacejka 的《Tire and Vehicle Dynamics》一书中。默认参数组取自该书，Pacejka 96 的参数也可以改写过来使用。Reference Load、Reference Camber 和 Reference Tire Pressure 这类参考参数只用于预览，不影响轮胎模型，也不会被保存。



Nominal Load : This value is also used in pacejka formula and here is a good explanation : https://www.tut.fi/ms/muo/vert/11_tyre_as_car_component/handling_braking_pure.htm

Nominal Load（名义载荷）：该值也用于 Pacejka 公式。这里有一份很好的说明：https://www.tut.fi/ms/muo/vert/11_tyre_as_car_component/handling_braking_pure.htm

Quoting from that source:

引用该来源：

The nominal tyre load is related to the maximum admissible static load for the specific temperature and speed

名义轮胎载荷与特定温度和速度下允许的最大静态载荷有关，

index, usually referred to as the ETRTO value(European Tyre and Rim Technical Organisation). The speed index indicates the maximum speed for which the tyre is allowed to be used, before it destroys itself due to overheating, as a result of high-frequency standing waves responsible for a strong increase of internal deformation power being converted into heat Choosing the nominal value Fz0 being equal to 80% of this ETRTO value, a reasonable choice for Fz0 is listed in table below:

该指标通常称为 ETRTO 值（欧洲轮胎轮辋技术组织）。速度级别表示轮胎允许使用的最高速度，超过后会因过热而损坏。过热来自高频驻波，它使内部变形功率大幅增加并转化为热量。把名义值 Fz0 选为该 ETRTO 值的 80%，下表列出了合理的 Fz0 选择：

Class 级别	**Fz0 [N]**

Fz0［牛］

Example
示例

Compactclass
紧凑级

3000
3000

VW-Polo
VW-Polo

Middleclass
中级

5000
5000

VW-Passat, BMW-5,..
VW-Passat、BMW-5 等

Topclass
高级

6000
6000

Audi A8
Audi A8

Hence, a specific nominal tyre load is related to a class of tyres, with the same maximum allowable operating speed. Different nominal tyre loads refer therefore to different classes of tyres, in contrast to the variation in tyre load for one specific tyre (due to static load variations, load transfer during cornering, etc.).
因此，某个名义轮胎载荷对应的是具有相同最高允许使用速度的一类轮胎。不同的名义轮胎载荷指的是不同级别的轮胎，这与某一条特定轮胎上的载荷变化不同（后者来自静态载荷变化、过弯时的载荷转移等）。

Nominal Tire Pressure : This value is required by the MF6.1 tire model for the pressure dependent effects. Normally this can be set to the default tire pressure(P) of the tire. Keep it between P - 0.5 and P + 0.5 bars.
Nominal Tire Pressure（名义胎压）：MF6.1 轮胎模型的压力相关效果需要该值。通常可以设为轮胎的默认胎压（P）。请保持在 P − 0.5 到 P + 0.5 巴之间。

Force Symmetry : If checked the negative side of the tire model curves will be symmetrical to the positive sides. Some tire model parameters can shift the origin of tire model curves and it might be desirable to avoid that but in most cases leave this disabled.
Force Symmetry（力对称）：若勾选，轮胎模型曲线的负侧会与正侧对称。有些轮胎模型参数会移动曲线原点，有时希望避免这一点，但多数情况下应保持禁用。